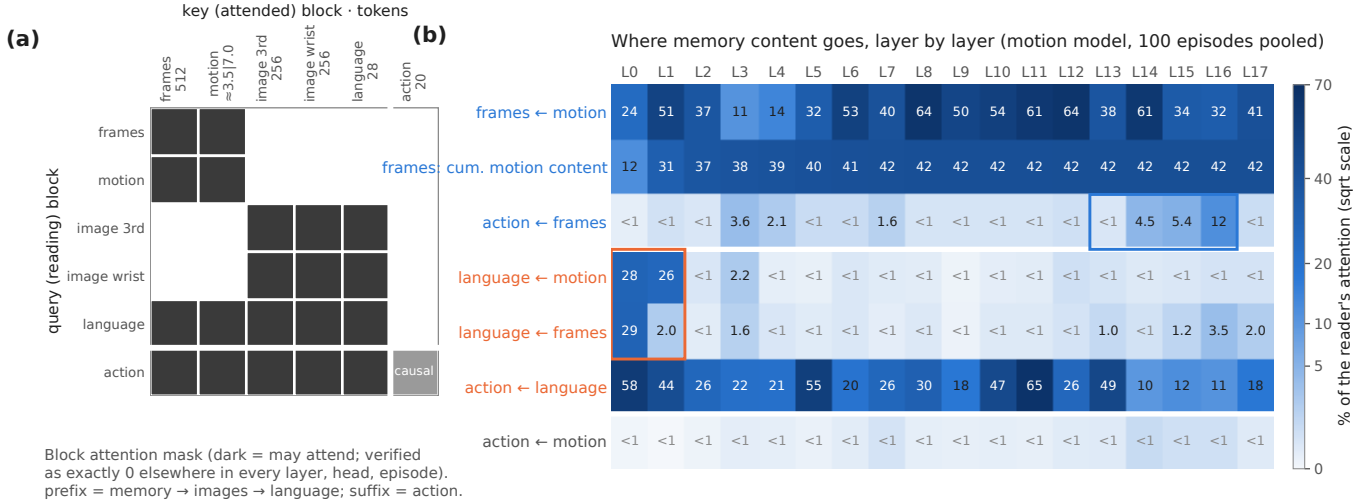
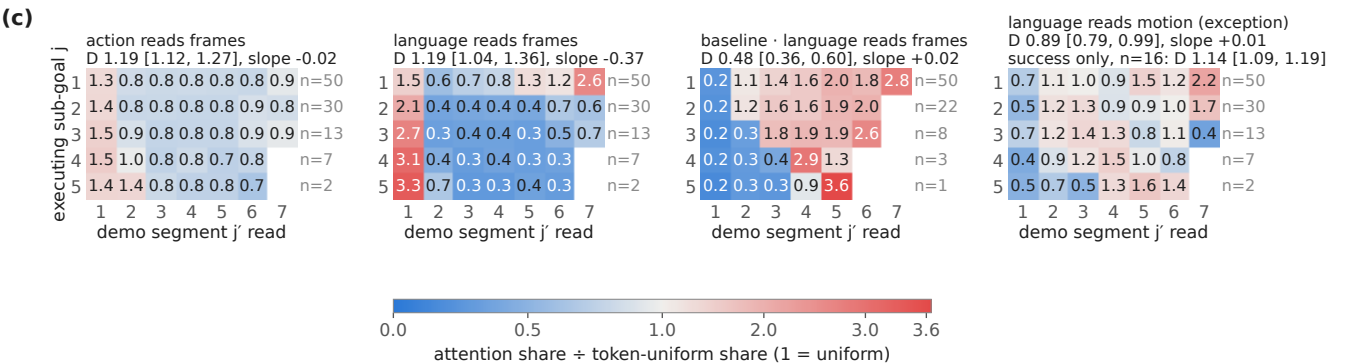
I. Two attention pathways carry motion-token content into the action tokens

Motion model, PatternLock | RouteStick, seed 42, 100 episodes / 495 inferences, head-averaged, episode-equal. (a) the attention mask as a block matrix; (b) per layer, how much language, action and frame memory read from motion tokens and frame memory (both tasks pooled).



Block attention mask (dark = may attend; verified as exactly 0 elsewhere in every layer, head, episode). prefix = memory → images → language; suffix = action.

Rows: 'reader ← source' = share of the reader's attention on the source at that layer; 'cum. motion content' = motion share of frame memory after that layer (rollout, α = 0.5). Blue labels: Route A via frame memory; orange: Route B via language. Outlined: memory → language only at L0–L1, frames → action only at L13–L16; action never reads motion directly (<1%). Colour scale is square-root.

II. The pathways exist, but which demo segment is read shows no sub-goal alignment

RouteStick, motion model unless noted, 50 episodes per model, all inferences; rows pool episodes with different numbers of sub-goals m (n = episodes per row); frame stream weighted 'single', motion stream 'overlap'; D = diagonal enrichment, episode-level 95% CI.